

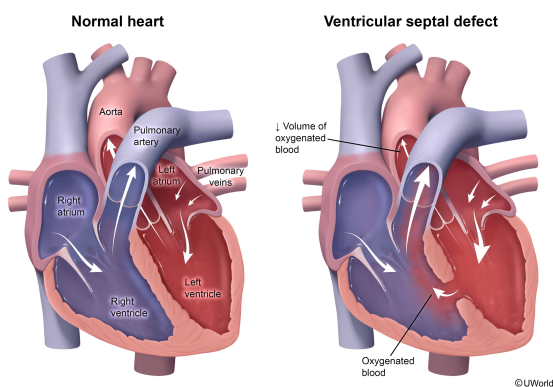
A male patient was born with an abnormal opening in the muscular wall separating the two ventricles of the heart. The opening allows blood to flow from the left to the right ventricle. If left untreated, this cardiac defect may cause any of the following in the patient EXCEPT:

- ☐ A. increased blood flow into the pulmonary artery. [11%]
- ☐ B. increased blood volume in the left atrium. [26%]
- ☐ C. low volume of oxygenated blood in the systemic arteries. [9%]
- ✓ ☒ D. reduced volume of oxygenated blood exiting the right ventricle. [52%]

Correct

51% answered correctly

Explanation:



The four chambers of the **heart** (two atria and two ventricles) pump blood throughout the **circulatory system**. Deoxygenated blood returning to the heart through systemic veins fills the right atrium and is pumped into the right ventricle. Contraction of the right ventricle drives blood through the pulmonary artery to the lungs via the pulmonary circuit. Oxygenated blood returning from the lungs via the pulmonary veins enters the left atrium and is pumped into the left ventricle, a highly muscular chamber that pumps oxygenated blood throughout the entire body via systemic arteries. Barriers called **septa**, composed of both **membranous and muscular tissue**, separate the two atria and two ventricles, preventing oxygenated and deoxygenated blood from mixing within the heart.

In this scenario, the patient was born with an abnormal opening in the muscular wall separating the ventricles of the heart (ie, **the ventricular septum**) that allows blood to flow from the left to the right ventricle. Because the left ventricle contains oxygenated blood, this opening in the ventricular septum causes **oxygenated blood to flow into the right ventricle**. If left untreated, this cardiac defect may cause:

- *Increased blood flow into the pulmonary artery.* Blood flow from the left to the right ventricle *increases* the volume of blood in the right ventricle and subsequently *increases* the amount of blood pumped through the pulmonary artery by right ventricular contraction (**Choice A**).
- *Increased blood volume in the left atrium.* Increased blood flow through the pulmonary circuit from the right ventricle likely *increases* the volume of blood flowing

through the pulmonary veins into the left atrium **(Choice B)**.

- *Low volume of oxygenated blood in the systemic arteries.* Because oxygenated blood abnormally flows from the left ventricle into the right ventricle and through the pulmonary circuit, the volume of oxygenated blood pumped from the left ventricle through systemic arteries decreases **(Choice C)**.

(Choice D) The abnormal flow of oxygenated blood into the right ventricle increases, *not* reduces, the volume of oxygenated blood exiting the right ventricle (ie, into the pulmonary artery).

Educational objective:

Deoxygenated blood returning to the heart enters the right atrium, is transferred to the right ventricle, and is pumped to the lungs via the pulmonary circuit. The left atrium receives oxygenated blood and transfers this blood to the left ventricle, which pumps oxygenated blood throughout the body.

Time spent:0 sec

Subject:
Biology

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System:
Circulation and Respiration